Mukund Telukunta | Curriculum Vitae

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Summary

AI Researcher specializing in **algorithmic fairness** and **responsible AI** within kidney transplantation, with expertise in developing, evaluating, and ensuring the ethical deployment of machine learning systems. Proficient in **PyTorch**, **TensorFlow**, and end-to-end ML workflows, with a proven track record of collaborating with diverse teams to deliver robust and fair AI solutions. Skilled in deep learning, reinforcement learning, Generative AI, and computer vision, with a focus on developing scalable solutions.

Education

Missouri University of Science and Technology — Rolla, MO
PhD in Computer Science — May 2025
Research Areas: Algorithmic Fairness, Machine Learning, Computer Vision

Missouri University of Science and Technology — Rolla, MO
Masters in Computer Science — December 2020

Technical Skills

Programming Languages: Python, R, SQL, Java, C/C++, JavaScript, HTML/CSS
Statistical & ML Tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, statsmodels, Keras
Experimentation & Analytics: A/B Testing, Causal Inference, Hypothesis Testing, Power Analysis, Bayesian Methods
Data Engineering & Databases: SQL, PostgreSQL, MongoDB, Hadoop, Spark, Hive, ETL Pipelines
Cloud & Deployment: AWS, GCP, Heroku, GitHub Actions (CI/CD), MLOps (W&B)
Frameworks/Libraries: React.js, Flask, Node.js, Flask-SocketIO

Relevant Courses

Machine Learning and AI: Introduction to Artificial Intelligence, Machine Learning for Computer Vision, Advanced Topics in Data Mining, Big Data and Cloud Computing

Theoretical Computer Science and Mathematics: Mathematics Probability and Statistics, Applied Graph Theory, Algorithmic Game Theory, Analysis of Algorithms

Systems and Applications: Database Systems, Advanced Bioinformatics

Publications (https://scholar.google.com/citations?user=iNSSyw8AAAAJ)

1. **Telukunta, M.**, & Nadendla, V. S. S. "A Difficulty in Evaluating Perceived Group Fairness using Non-Expert Opinions." (In preparation.)
2. **Telukunta, M.**, Stuart, M., Nadendla, V. S. S., & Canfield, C. "Fairness Perception of Regression-based Predictive Analytics Tools used in Kidney Placement." Under Review. *Preprint*
3. **Telukunta, M.**, Rao, S., Stickney, G., Nadendla, V. S. S., & Canfield, C. (2024). "Learning Social Fairness Preferences from Non-Expert Stakeholder Opinions in Kidney Placement." *Proceedings of the 5th Conference on Health, Inference, and Learning (CHIL). Full PDF*
4. Sanga, S., **Telukunta, M.**, Nadendla, V. S. S., & Das, S. K. (2023). "Strategic Information Design in Selfish Routing with Quantum Response Travelers." *IEEE 20th International Conference on Mobile Ad Hoc Smart Systems (MASS). Full PDF*
5. **Telukunta, M.**, & Nadendla, V. S. S. (2023) "Towards Inclusive Fairness Evaluation via Eliciting Disagreement Feedback from Non-Expert Stakeholders." *3rd Workshop on Bias and Fairness in AI at European Conference on Machine Learning (ECML PKDD). Full PDF*

6. **Telukunta, M.**, & Nadendla, V. S. S. (2020) "On the Identification of Fair Auditors to Evaluate Recommender Systems based on a Novel Non-Comparative Fairness Notion." *3rd FAccTRec Workshop on Responsible Computing at ACM Recommender Systems (RecSys).* Full PDF

Experience

Missouri University of Science and Technology — Rolla, MO

Graduate Research Assistant - Data Science — August 2020 - Current

- Led experimentation design and analysis for research projects evaluating algorithmic fairness and causal inference across healthcare and consumer platforms, collaborating with cross-functional teams including clinicians, data scientists, and product stakeholders to define metrics and design scalable experiments.
- Designed and executed 15+ A/B tests and quasi-experimental studies using randomized controlled trials, difference-in-differences, and synthetic control methods to evaluate causal impact of ML models on decision-making outcomes. Leveraged Python, R, and SQL for data manipulation, statistical analysis, and metric computation.
- Established experimentation best practices and evaluation frameworks for assessing fairness and usability of predictive analytics tools, teaching methodologies to research teams and ensuring data-driven decision-making at scale across 500+ experiment participants.
- Built robust statistical models including Gradient Boosting, Random Forests, and Logistic Regression using Python (scikit-learn, XGBoost) and R to predict user behavior and preferences, achieving 92% prediction accuracy and providing actionable insights for product optimization.
- Conducted advanced data exploration and causal inference analyses on heterogeneous datasets using dimensionality reduction (PCA, t-SNE), outlier detection, and stratified sampling. Preprocessed and analyzed time-series and cross-sectional data from 10,000+ user interactions using SQL and Pandas.
- Acted as thought leader in experimentation and causal inference for healthcare applications, advising Product and Engineering teams on best practices for experiment design, metric selection, and statistical significance testing. Made compelling cases for prioritization and resource investments resulting in 3 funded research publications.
- Delivered high-quality analytical solutions through data modeling, machine learning, and statistical inference with emphasis on impactful results communicated to diverse audiences. Published 5 peer-reviewed papers in top-tier ML/AI conferences and workshops.
- Designed and deployed large-scale survey experiments on Prolific crowdsourcing platform using Power Analysis to determine optimal sample sizes, achieving statistical power of 0.85+ and collecting 5,000+ high-quality responses for data-driven product insights.
- Mentored 2 undergraduate researchers on experimentation methodologies and statistical analysis, fostering collaborative learning and enabling team members to independently design experiments and analyze results.
- Developed automated data pipelines and reproducible analytics workflows using Python, SQL, and GitHub, reducing analysis time by 60% and ensuring reproducibility across experimental iterations.

AT&T Labs — Des Peres, MO

Student Technical Intern - Data Analytics — June - August 2019

- Architected cloud-based reporting and analytics system using SQL, PHP, JavaScript, and AWS cloud platforms to deliver real-time financial performance dashboards for data-driven decision-making across 800-person organization.
- Engineered automated ETL pipelines for connecting and processing large datasets from relational (RDBMS) and non-relational (NoSQL) databases, reducing report generation time by 85% and improving forecasting accuracy by 30%.
- Analyzed product usage patterns and user behavior metrics using SQL queries and statistical methods, enabling product teams to identify optimization opportunities and prioritize feature investments.
- Collaborated with cross-functional stakeholders including product managers, engineers, and business analysts to translate analytical requirements into technical solutions, achieving 95% user adoption rate within 6 months.

- Delivered data insights and visualizations to non-technical audiences, effectively communicating complex analytical findings to drive product and operational decisions.

Projects (https://github.com/mukund0911)

WaveCrafter: AI-Powered Multi-Speaker Voice Editor | *Try it:* https://wave-crafter.com/

- Built full-stack ML platform using **React**, **Flask**, **PyTorch**, and **VoiceCraft (330M)** model. Designed and executed A/B experiments to evaluate voice cloning quality across different inference modes, achieving **95%+ voice similarity** and validating product performance using statistical hypothesis testing.
- Conducted causal analysis on voice cloning quality improvements by implementing **Montreal Forced Aligner (MFA)** for word-level alignment, measuring **35-point improvement (60% to 95%+)** in similarity scores through controlled experiments comparing baseline vs. MFA-enhanced models.
- Optimized system performance through experimentation-driven approach: reduced latency from 15-18s to **6-8 seconds** (63% improvement) via container warmup experiments and achieved **3x speedup** through parallel processing A/B tests on serverless **AWS A10G GPU infrastructure**.
- Developed composite quality scoring system using 6 statistical metrics (**SNR, RMS energy, spectral richness**) validated through offline experiments. Built ML classifier using **scikit-learn** with feature engineering (MFCC extraction) processing **30-min podcasts in 46 seconds**.
- Architected scalable cloud solution using **AWS S3**, **Heroku**, **Modal.com**, and **AssemblyAI API**, reducing operational costs by **90% ($350 to $33/month)** through data-driven resource optimization and automated deployment via **GitHub Actions CI/CD**.

Experimentation Platform for Fairness Evaluation in Healthcare ML Models

- Designed and implemented scalable experimentation framework to evaluate fairness perceptions of ML-based kidney placement algorithms, establishing metrics and statistical methodologies for assessing algorithmic fairness across diverse stakeholder groups.
- Conducted 3 large-scale randomized experiments (N=500+) using difference-in-differences and within-subjects designs to measure causal impact of algorithm explanations on fairness perceptions. Performed statistical analysis using **Python** and **R** including ANOVA, t-tests, and regression modeling.
- Built data collection and experimentation infrastructure on Prolific platform using **RESTful APIs**, **SQL databases**, and automated data pipelines. Implemented quality controls and real-time monitoring reducing invalid responses by 40%.
- Defined and tracked 15+ evaluation metrics including fairness ratings, decision confidence, and user satisfaction. Performed exploratory data analysis, feature engineering, and dimensionality reduction to identify key drivers of fairness perceptions.
- Delivered actionable insights to cross-functional stakeholders including clinicians, procurement organizations, and policy makers through data visualization, statistical reports, and compelling presentations. Published findings in 3 peer-reviewed conferences.
- Applied advanced causal inference techniques including propensity score matching, instrumental variables, and mediation analysis to establish causal relationships between algorithm features and user perceptions, informing product design decisions.

DNABART: Genomic LLM for Sequence Correction and Classification

- Developed large-scale machine learning models using **BART encoder-decoder** architecture and **PyTorch** for genomic sequence analysis. Designed offline experiments to evaluate model performance across 26 benchmark tasks, achieving state-of-the-art results on 16 tasks with **up to 24% accuracy improvement**.
- Implemented experimentation workflows using **Weights & Biases (W&B)** for hyperparameter tuning, metric tracking, and A/B comparisons of model architectures, preprocessing strategies, and optimization techniques across NVIDIA H100/V100 GPU infrastructure.
- Conducted systematic evaluation of tokenization strategies (**Byte-Pair Encoding**), pretraining objectives, and fine-tuning approaches through controlled experiments on **15GB DNABERT2** and **1M sequence datasets**, identifying optimal configurations for model convergence and generalization.
- Built scalable data pipelines and reproducible ML workflows for large-scale genomic data processing,

reducing training time by 40% through compute optimization and efficient data loading strategies on HPC clusters.

Real-time Multi-Sensor Multi-Object Detection via Transfer Learning

- Fine-tuned the YOLOv6 object detection model using transfer learning to detect NVIDIA Jetson Robots on a custom-built testbed, achieving a detection accuracy of 98.5%.
- Designed and developed a real-time multi-sensor vision system by integrating images from multiple camera sensors. Leveraged OpenCV Stitcher class to create seamless panoramic views and implemented advanced detection algorithms like SIFT (Scale-Invariant Feature Transform) for enhanced feature extraction.
- Implemented Kalman Filtering for robust real-time tracking of Jetbot robots, including state vector estimation and covariance matrix updates. Achieved a tracking accuracy of 95% under dynamic conditions, with precise object state prediction and noise filtering.
- Engineered a multi-camera localization system by positioning multiple sensors as satellites around the testbed to triangulate and determine robot positions. This ongoing effort aims to improve localization accuracy by 20%, enabling sub-centimeter precision for real-time robot tracking.
- Optimized system pipeline for real-time performance using OpenCV, Kalman Filtering, and YOLOv6 on NVIDIA Jetson hardware, ensuring low-latency operation and computational efficiency.

COVID-19 Detection in Chest X-Rays Using Custom CNN and Grad-CAM

- Designed and implemented a custom Convolutional Neural Network (CNN) to predict COVID-19 infections from chest X-ray images using TensorFlow and Keras, achieving an accuracy of 80% on a limited dataset.
- Enhanced model interpretability by implementing Grad-CAM (Gradient-weighted Class Activation Mapping) to visualize critical regions in X-ray images that influenced predictions, enabling clearer clinical insights for healthcare professionals.
- Improved model generalization on the small dataset through extensive data augmentation and transfer learning using pre-trained models like VGG16 and ResNet50, boosting performance metrics by 15% compared to the baseline CNN model.
- Optimized preprocessing pipelines using OpenCV for image resizing, normalization, and augmentation (e.g., rotation, flipping, and contrast adjustment) to enhance dataset variability and reduce overfitting.
- Evaluated model robustness using cross-validation techniques and tracked performance metrics such as accuracy, precision, recall, and F1-score for thorough assessment.

Teaching

Introduction to C++ Programming (CS1580) :

- Taught undergraduate students foundational C++ programming concepts, including object-oriented programming (OOP), data types, control structures, and functions, building a strong programming foundation.
- Designed and conducted hands-on programming assignments and projects that introduced concepts like classes, inheritance, and polymorphism to encourage problem-solving and logical thinking.
- Held weekly coding labs and office hours to provide individualized guidance, clarify concepts, and troubleshoot programming errors.

Data Structures (CS1585) :

- Instructed undergraduates on essential software development tools, including version control systems (e.g., Git), unit testing frameworks, and regular expressions (RegEx), equipping students with practical industry skills.

Machine Learning in Computer Vision (CS6406) :

- Assisted the course instructor in designing assignments integrating PyTorch and TensorFlow frameworks, covering topics like model training, evaluation, and deployment in real-world vision tasks.
- Delivered recitation lectures on advanced topics such as CUDA Programming for GPU acceleration, Convolutional Neural Networks (ResNet, VGGNet), Optimization Algorithms (SGD, Adam), and Transfer Learning, providing both theoretical depth and hands-on implementation examples.

- Created supplementary materials, including cheat sheets, code repositories, and recorded lectures

Invited Panels & Awards

Bridging Disparities in Health Care using AI Symposium	November 2024
AI and You Symposium Panel at the Center for Science, Technology and Society	May 2024
Participation in Intelligent Systems Center Poster Presentation	October 2023
Runner-up IEEE St.Louis Student Presentations	April 2022
1st Place in Graduate Research Poster Contest	May 2019